

Ww2

Code:

lijn Tilburg - Breda

125 km 24.343

Overbragt : Mortelbestraat.

Soort : vaste, enkelsporige brug.

Type : dubbelrijzer

Wijdte in de dag: , m



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Theoretische lengte hoofdliggers: 6,72 m = L

Afstand hoofdliggers h.o.h. : 0,700 - 0,815 - 0,700 m

Afstand dwarsdragers h.o.h. : , m = l

Maximum belasting hoofdliggers: $\frac{14,0}{16,5}$ t/m' spoor bij een lengte L = 6,72 m

" " dwarsdragers: $\frac{14,0}{16,5}$ t/m' " " " " 2 l = m

" " langdragers: $\frac{14,0}{16,5}$ t/m' " " " " l = m

Materiaal: Staal

Berekend volgens : ROB 1971

Gemaakt volgens bestek: RWS 115 (dienstjaar 1918) Bouwjaar: 1920

Tekeningen :

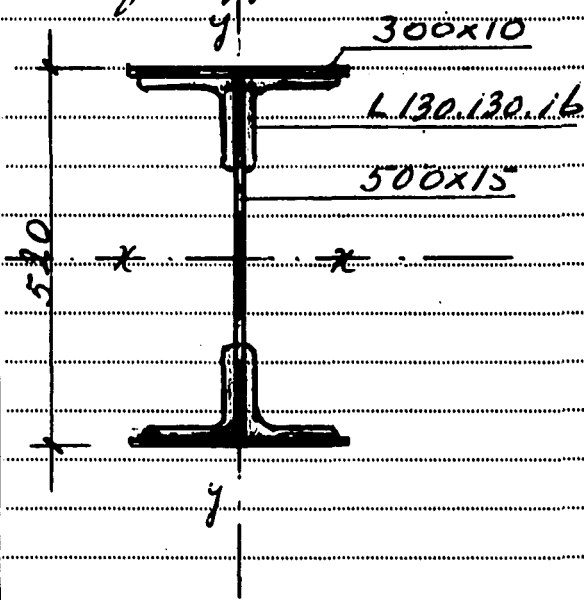
Omschrijving:

Opmerkingen:

Op brug toe te laten NBV 1917

MP 22/4/87.

Houtfalsligger



$$l_{th} = 6,72m$$

$$eigen\ gewicht = 1,40 \text{ t/m'}$$

$$nagels: \phi 20$$

$$steek: 80$$

$$L 130.130.16$$

$$e = 3,80cm; A = 39,3cm^2;$$

$$I_x = 605cm^4$$

$$A_{profiel}: 50,0 \times 1,5 + 4 \times 39,3 + 2 \times 30,0 \times 1,0 = 292,2cm^2$$

$$\sigma_{stat} = 140 N/mm^2$$

$$\tau_{stat} = 81 N/mm^2$$

$$schootcoefficient: 1,12 + \frac{2,5}{6,72} = 1,49$$

I_x :

$$\frac{1}{12} \times 1,5 \times 50,0^3 = 15.625cm^4$$

$$4 \times 605 = 2.420 "$$

$$4 \times 39,3 \times 21,2^2 = 70.652 "$$

$$2 \times \frac{1}{12} \times 30,0 \times 1,0^3 = 5 "$$

$$2 \times 1,0 \times 30,0 \times 25,5^2 = 39.015 "$$

$$127.717cm^4$$

S_x :

$$1,0 \times 30,0 \times 25,5 = 765cm^3$$

$$2 \times 39,3 \times 21,2 = 1.666 "$$

$$S_{\overline{x}} = 2431cm^3$$

$$\frac{1}{2} \times 25,0^2 \times 1,5 = 469 "$$

$$S_x = 2900cm^3$$

$$W_{zbruto} = \frac{127.717}{26,0} = 4912cm^3 \quad W_{xnetto} = 0,8 \times W_{zbruto} = 3930cm^3$$

$$q_{mC4} \text{ bij } l_g 6,72m : 95 kN/m'$$

$$q_{vC4} \text{ bij } l_g 6,72m : 123 kN/m'$$

Momenten

$$M_{eg}: 10 \times 14,0 \times 6,72^2 = 79 kNm$$

$$M_{mib}: 10 \times 95,16 \times 6,72^2 \times 1,49 = 799 "$$

$$M_{max} = 878 kNm; M_{min} = 79 kNm$$

$$\sigma_{eg} = \sigma_{min} = \frac{79 \times 10^3}{4 \times 3930} = 5 N/mm^2$$

$$\sigma_{max} = \frac{878 \times 10^3}{4 \times 3930} = 56 N/mm^2$$

$$\sigma_{mib} = 51 N/mm^2$$

$$\bar{q}_m = 2,64 \times 95 = 249 kN/m'$$

$$N_{DU1917} : 140 kN/m'$$

$$Toe \text{ te laten: } \frac{140 - 5}{31} \times 100 = 264 \% C4$$

Doorbuiging ($\bar{f} = 800 \text{ l}$)

$$q_m = \frac{0,2016 \times 4 \times 127717}{6,72^3} = 339 \text{ kN/m} > \text{NBV 1917: } 140 \text{ kN/m}$$

Toe te laten: $\frac{339}{95} \times 100 = \underline{\underline{357 \% C4}}$

Dwarskrachten

$$D_{eg} = \frac{1}{2} \times 14,0 \times 6,72 = 47 \text{ kN}$$

$$D_{mob} = \frac{1}{2} \times 123 \times 6,72 \times 1,49 = 616$$

$$D_{max} = 663 \text{ kN}; D_{min} = 47 \text{ kN}$$

$$I_{eg} = I_{min} = \frac{47 \times 2900 \times 10^4}{4 \times 1,5 \times 127717} = 2 \text{ N/mm}^2$$

$$I_{max} = \frac{663 \times 2900 \times 10^4}{4 \times 1,5 \times 127717} = 25 \text{ N/mm}^2$$

$$I_{mob} = 23 \text{ N/mm}^2$$

Toe te laten $\frac{81-2}{23} \times 100 = \underline{\underline{343 \% C4}} \quad \bar{q}_v = 3,43 \times 123 = 422 \text{ kN/m}$

Nagels in randbalkstalen

$$> \text{NBV 1917: } 165 \text{ kN/m}$$

$$K_{eg} = \frac{47 \times 2431 \times 8,0}{4 \times 127717} = 2 \text{ kN}$$

$$K_{max} = \frac{663 \times 2431 \times 8,0}{4 \times 127717} = 25 \text{ kN}$$

$$K_{mob} = 23 \text{ kN}$$

$\bar{K}$ nagel $\phi 20$ $\delta = 15$ dubbelmedig: 70 kN

Toe te laten: $\frac{70-2}{23} \times 100 = \underline{\underline{295 \% C4}}$

$$\bar{q}_v = 2,95 \times 123 = 362 \text{ kN/m} > \text{NBV 1917: } 165 \text{ kN/m}$$

Overzicht procentuele toelaatbare belasting C4 op de brug

		ligger
moment	$\bar{G}$	264
doorbuiging		357
dwarskracht	$\bar{I}$	343
nagels		295

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